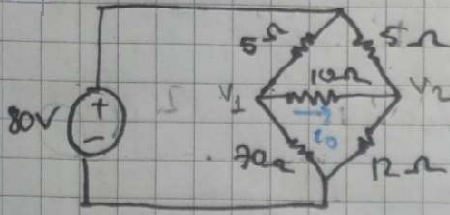


$$i_0 = ?$$



$$\frac{V_1}{70} + \frac{V_1 - V_2}{10} + \frac{V_1 - 80}{5} = 0$$

$$\frac{V_2}{12} + \frac{V_2 - V_1}{10} + \frac{V_2 - 80}{5} = 0$$

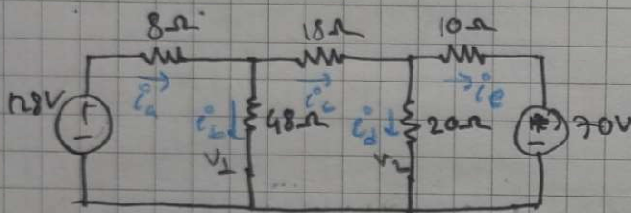
$$22V_1 - 7V_2 = 1120$$

$$-6V_1 + 23V_2 = 960$$

$$V_1 = 70V \quad V_2 = 60V$$

$$i_0 = \frac{V_1 - V_2}{10} = \underline{\underline{1A}}$$

$$i_a, i_b, i_c, i_d = ?$$



$$\frac{V_1 - 128}{8} + \frac{V_1}{48} + \frac{V_1 - V_2}{18} = 0$$

$$\frac{V_2 - 70}{10} + \frac{V_2}{20} + \frac{V_2 - V_1}{18} = 0$$

$$V_1 \left(\frac{1}{8} + \frac{1}{48} + \frac{1}{18} \right) + V_2 \left(\frac{-1}{18} \right) = \frac{128}{8}$$

$$V_1 \left(\frac{-1}{18} \right) + V_2 \left(\frac{1}{18} + \frac{1}{20} + \frac{1}{10} \right) = \frac{70}{18}$$

$$\left. \begin{array}{l} V_1 = 86V \\ V_2 = 60V \end{array} \right\}$$

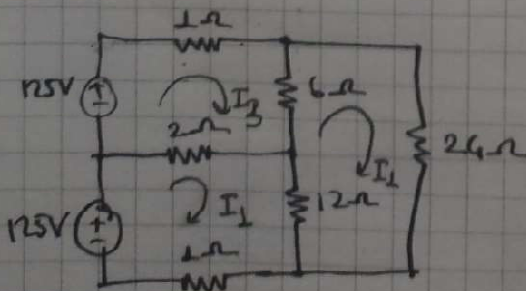
$$i_a = (128 - 86) / 8 = \underline{\underline{4A}}$$

$$i_b = 86 / 48 = \underline{\underline{2A}}$$

$$i_c = (86 - 60) / 18 = \underline{\underline{2A}}$$

$$i_d = 60 / 20 = \underline{\underline{3A}}$$

göz akımlarını bulunuz.



$$42I_1 - 12I_2 - 2I_3 = 0$$

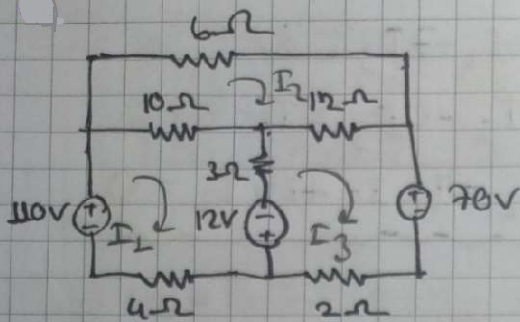
$$-6I_1 - 2I_2 + 9I_3 = 125$$

$$-12I_1 + 15I_2 - 2I_3 = 125$$

$$I_1 \approx \underline{\underline{5,41A}}$$

$$I_2 \approx \underline{\underline{15,43A}}$$

$$I_3 \approx \underline{\underline{20,93A}}$$



$$\begin{aligned} 17I_1 - 10I_2 - 3I_3 &= 110 + 70 \\ -10I_1 + 28I_2 - 12I_3 &= 0 \\ -3I_1 - 12I_2 + 12I_3 &= -82 \end{aligned}$$

$$I_1 = 8A \quad I_2 = 2A \quad I_3 = -2A$$

a) toplam güç?

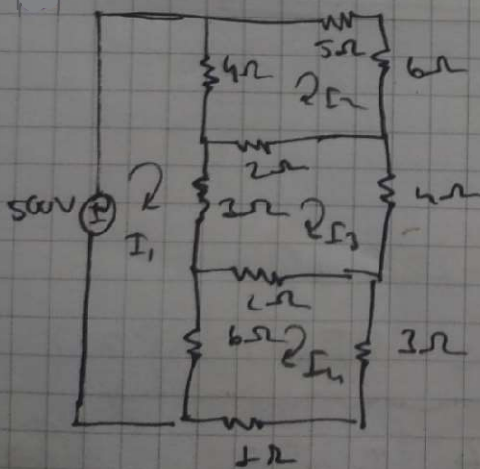
$$\begin{aligned} P_{110} &= 110I_1 = 880W \\ P_{12} &= 12(I_1 - I_2) = 120W \\ P_{70} &= -70I_3 = 140W \end{aligned}$$

$$P_{toplam} = 1140W$$

b) üretilen P=? harcanan

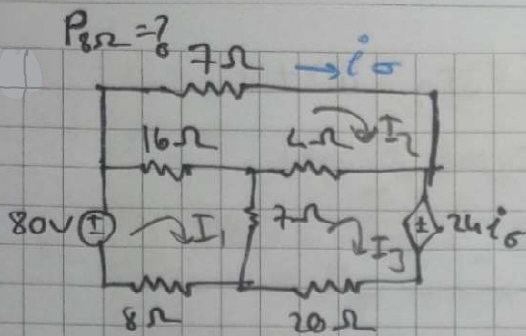
$$\begin{aligned} P_G &= 1140W \text{ (a şıkkı)} \\ P_{4\Omega} &= 8^2 \cdot 4 = 256W & P_{2\Omega} &= (-2)^2 \cdot 2 = 8W \\ P_{6\Omega} &= 6^2 \cdot 10 = 360W & P_{0\Omega} &= 2^2 \cdot 6 = 24W \\ P_{2\Omega} &= (-4)^2 \cdot 12 = 192W & P_{3\Omega} &= 10^2 \cdot 3 = 300W \end{aligned}$$

$$P_{Harcana} = 1140W$$



$$\begin{aligned} 13I_1 - 4I_2 - 3I_3 - 6I_4 &= 500 \\ -4I_1 + 13I_2 - 2I_3 &= 0 \\ -3I_1 - 2I_2 + 11I_3 - 2I_4 &= 0 \\ -6I_1 - 2I_3 + 12I_4 &= 0 \end{aligned}$$

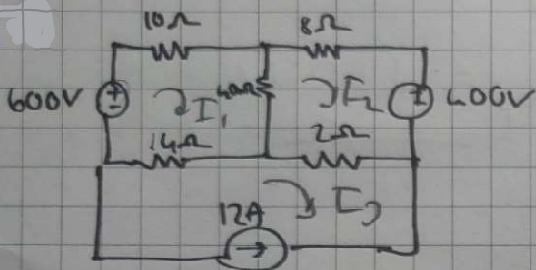
$$\begin{aligned} I_1 &= 74.48 \\ I_2 &= 28.02 \\ I_3 &= 33.18 \\ I_4 &= 42.77 \end{aligned}$$



$$P_{8\Omega} = (3,5)^2 \cdot 8 = \underline{\underline{98W}}$$

$$\begin{aligned} 3I_1 - 16I_2 - 7I_3 &= 80 \\ -16I_1 + 27I_2 - 4I_3 &= 0 \\ -7I_1 - 4I_2 + 18I_3 + 2I_2 &= 0 \end{aligned}$$

$$I_1 = 3,5A$$



$$\begin{aligned} 64I_1 - 40I_2 - 14I_3 &= 600 \\ -40I_1 + 50I_2 - 2I_3 &= -400 \\ -12 &= I_3 \end{aligned}$$

$$I_1 = 2,9A \quad I_2 = -6,16A \quad I_3 = -12A$$

a) $P_{12A} = I_3 \cdot V = \underline{\underline{2643,36W}}$
 $I = 12A$

$$\begin{aligned} V &= 12(12 - 6,16) + 14(12 + 2,9) \\ &= 270,28V \end{aligned}$$

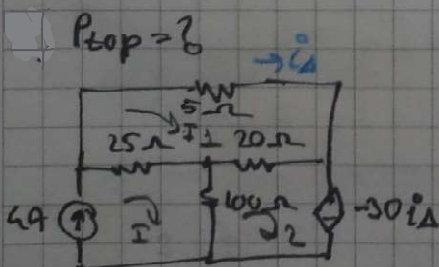
b) $P_{600V} = 600 \cdot (-6,16) = 2464W$

$$P_{600V} = 600 \cdot 2,9 = 1740W$$

$$P_{12A} = (12 \cdot 12) = 2643,36W$$

$$P_{\text{total}} = 6847,36W$$

c) $(2,9)^2 \cdot 10 + (6,16)^2 \cdot 8 + (3,06)^2 \cdot 40 +$
 $(4,9)^2 \cdot 14 + (5,84)^2 \cdot 2 = \underline{\underline{6847,36W}}$



$$\begin{aligned} 50I_1 - 20I_2 - 25I_3 &= 0 \\ -20I_1 + 120I_2 - 30I_3 - 100I &= 0 \end{aligned}$$

$$I = 4A \quad I_1 = 4A \quad I_2 = 5A \quad I_3 = I_1$$

$$P_{30\Omega} = (30,1) \cdot I_2 = \underline{\underline{600W}}$$



$$18I_1 - 16I_2 = 10$$

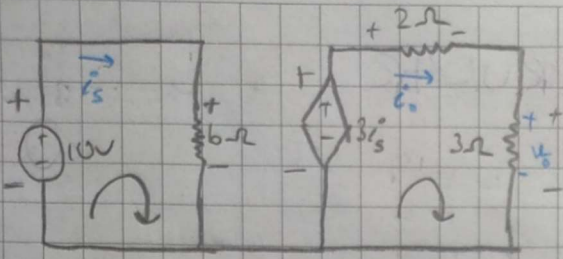
$$-16I_1 + 28I_2 + 4I_0 = 0$$

$$4 = 8I_0$$

$$I_1 = 1A \quad I_2 = 0.5A \quad I_0 = 0.5A$$

a) $V_0 = 16(0.5) = \underline{\underline{8V}}$

b) $P_{4I_0} = 4(0.5)(0.5) = \underline{\underline{1W}}$



$$-10V + 6\Omega \cdot i_s = 0$$

$$-3i_s + 2\Omega i_o + 3\Omega i_o$$

$$10 = 6i_s$$

$$3i_s = 5i_o$$

$$i_s = 5/3 A \approx 1.66 A$$

$$i_o = 1 A$$

$$a) V_o = 1 \cdot 3 = 3V$$

$$b) \frac{25}{3}$$

$$b) p = V \cdot I = I^2 \cdot R$$

Üretilen

$$p = 10 \cdot \frac{5}{3} + 3 \cdot \frac{5}{3} \cdot 1$$

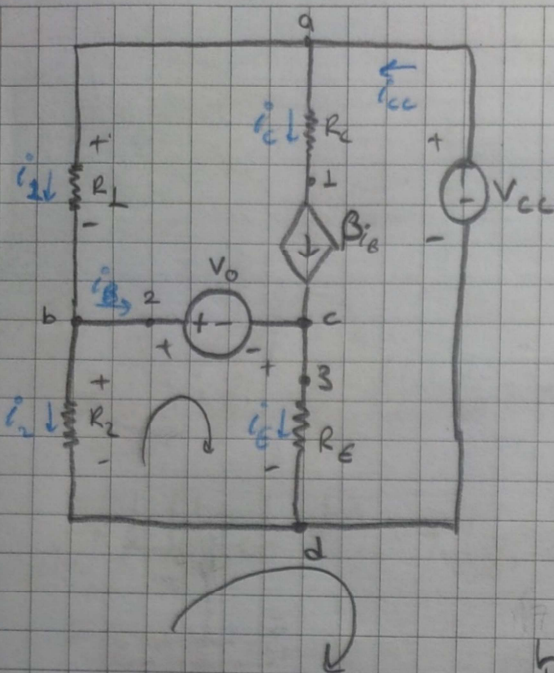
$$\approx 21.66 W$$

Harcanan

$$p = 6 \cdot \left(\frac{5}{3}\right)^2 + 2 \cdot 1^2 + 3 \cdot 1^2$$

$$\approx 21.66 W$$

harcanan ve üretilen güçler eşittir.



a)

$$i_{CC} = i_C + i_L$$

$$i_C = \beta i_B$$

$$i_L = i_B + i_C$$

$$i_E = i_C + i_B$$

$$V_o + i_E R_E - i_2 R_2 = 0$$

$$V_{CC} - i_2 R_2 - i_1 R_1 = 0$$

b) ?